Duality

Lecture 5

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Reference Reading:

Bertsimas and Tsitsiklis, *Introduction to Linear Optimization*, Chapters 4.1-4.4.

Winston, *Operations Research*, Chapter 6.

Vanderbei, *Linear Programming*, Chapter 5.

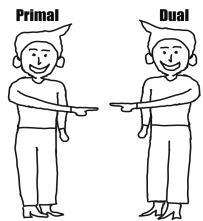
Lecture Overview



- The dual problem of an LP
- Weak and strong duality
- Complementary slackness
- Variants and examples
- Sensitivity analysis

The Dual Problem

The dual problem acts like a twin problem to an optimization problem.



Motivation

Consider an LP:

$$\begin{aligned} \text{maximize} \quad & 4x_1 + x_2 + 3x_3 \\ \text{subject to} \quad & x_1 + 4x_2 \leq 1 \\ & 3x_1 - x_2 + x_3 \leq 3 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Observe that every feasible solution provides a lower bound on $p^\star$, for example:

- $x = (1, 0, 0)$ tells us that $p^\star \geq 4$
- $x = (0, 0, 3)$ tells us that $p^\star \geq 9$

But how good are these bounds? Are they close to optimal value?

To answer, we need to provide upper bounds.

Upper bounds on the optimal objective value

Let's multiply the first constraint by 2 and add that to 3 times the second constraint:

$$\begin{array}{rl} 2(x_1 + 4x_2) & \leq 2(1) \\ +3(3x_1 - x_2 + x_3) & \leq 3(3) \\ \hline 11x_1 + 5x_2 + 3x_3 & \leq 11 \end{array}$$

since each variable is nonnegative, we have:

$$4x_1 + x_2 + 3x_3 \leq 11x_1 + 5x_2 + 3x_3 \leq 11$$

Hence, $p^\star \leq 11$.

Combining with the result from previous slide, we have:

$$9 \leq p^\star \leq 11$$

These bounds can be improved.

Improving the bounds

We apply the same upper bounding technique, but replace the specific numbers we used before with variables:

$$\begin{array}{rl} y_1(x_1 + 4x_2) & \leq y_1 \\ +y_2(3x_1 - x_2 + x_3) & \leq 3y_2 \\ \hline (y_1 + 3y_2)x_1 + (4y_1 - y_2)x_2 + (y_2)x_3 & \leq y_1 + 3y_2 \end{array}$$

If we stipulate that each of the coefficients of the x_i's be at least as large as the corresponding coefficient in the objective function,

$$\begin{aligned} y_1 + 3y_2 &\geq 4 \\ 4y_1 - y_2 &\geq 1 \\ y_2 &\geq 3 \end{aligned}$$

we can compare the objective function against this sum (and its bound):

$$\begin{aligned} p &= 4x_1 + x_2 + 3x_3 \\ &\leq (y_1 + 3y_2)x_1 + (4y_1 - y_2)x_2 + (y_2)x_3 \\ &\leq y_1 + 3y_2 \end{aligned}$$

Getting the best upper bound

We now have an upper bound, $y_1 + 3y_2$, which we should minimise in our effort to obtain the best possible upper bound:

$$\begin{aligned} \text{minimize} \quad & y_1 + 3y_2 \\ \text{subject to} \quad & y_1 + 3y_2 \geq 4 \\ & 4y_1 - y_2 \geq 1 \\ & y_2 \geq 3 \\ & y_1, y_2 \geq 0 \end{aligned}$$

This problem is called the **dual linear programming problem** associated with the given linear programming problem.

Dual of LP in canonical form

Given an LP in canonical form with parameters $c \in \mathrm{R}^n$, $A \in \mathrm{R}^{m \times n}$, $b \in \mathrm{R}^m$

$$\begin{array}{ll} \text{maximize} & c^T x \\ \text{subject to} & Ax \leq b \\ & x \geq 0 \end{array} \tag{1}$$

its **dual** is

$$\begin{array}{ll} \text{minimize} & b^T y \\ \text{subject to} & A^T y \geq c \\ & y \geq 0 \end{array} \tag{2}$$

in the context of duality, the LP (1) is called the **primal** problem

The dual of the dual

Let's show that taking the dual of the dual returns us to the primal.

First, convert the dual to canonical form:

$$\begin{aligned} \text{minimize} \quad & b^T y \\ \text{subject to} \quad & A^T y \geq c \\ & y \geq 0 \end{aligned} \quad \Longrightarrow \quad \begin{aligned} \text{maximize} \quad & -b^T y \\ \text{subject to} \quad & -A^T y \leq -c \\ & y \geq 0 \end{aligned}$$

Now let's take the dual and convert it to canonical form:

$$\begin{aligned} \text{minimize} \quad & -c^T x \\ \text{subject to} \quad & -(A^T)^T x \geq -b \\ & x \geq 0 \end{aligned} \quad \Longrightarrow \quad \begin{aligned} \text{maximize} \quad & c^T x \\ \text{subject to} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

Weak and strong duality

With the help of duality, we can obtain very helpful bounds on the primal objective value based on the dual objective value. Note that this concept of duality extends beyond linear programming and applies to other class of mathematical optimization.

Weak duality

Theorem (weak duality)

Consider maximization LP in canonical form. If $\tilde{x}$ is primal feasible and $\tilde{y}$ is dual feasible, then

$$c^T \tilde{x} \leq b^T \tilde{y}$$

Proof.

$$c^T \tilde{x} \leq \tilde{y}^T A \tilde{x} \leq \tilde{y}^T b$$

The first inequality follows from $x \geq 0$ and $c \leq y^T A$. The second inequality follows from $y \geq 0$ and $Ax \leq b$. □

Weak duality

Corollary

If an LP and its dual are both feasible, then both are bounded feasible.

If the primal is unbounded, then the dual must be infeasible.

If the dual is unbounded, then the primal must be infeasible.

Strong duality

Corollary

If there exist feasible $x^\star$ and $y^\star$ for LP (1) *and its dual* (2) *such that $c^T x^\star = b^T y^\star$, then both $x^\star$ and $y^\star$ are optimal for their respective problems.*

Proof.

If $\tilde{x}$ is any feasible vector for (1), then

$$c^T \tilde{x} \leq b^T y^\star = c^T x^\star$$

which shows that $x^\star$ is optimal. A symmetric argument works for $y^\star$. □

Theorem (The duality theorem)

If LP (1) *is bounded feasible, then so is its dual, their values are equal ($p^\star = d^\star$), and there exist optimal vectors for both problems.*

Example: Dakota furniture company

The Dakota Furniture company manufactures desk, tables, and chairs. The manufacture of each type of furniture requires lumber and two types of skilled labor: finishing and carpentry. The amount of each resource needed to make each type of furniture is given in the table below.

Currently, 48 board feet of lumber, 20 finishing hours, 8 carpentry hours are available. A desk sells for $60, a table for $30, and a chair for $20. Dakota believes that demand for desks and chairs is unlimited, but at most 5 tables can be sold. Because the available resources have already been purchased, Dakota wants to maximize total revenue.

resource	desk	table	chair
lumber (board feet)	8	6	1
finishing (hours)	4	2	1.5
carpentry (hours)	2	1.5	0.5

Example: Dakota furniture company

primal problem:

$$\begin{array}{rl} \text{maximize} & 60x_1 + 30x_2 + 20x_3 \\ \text{subject to} & 8x_1 + 6x_2 + x_3 \leq 48 \\ & 4x_1 + 2x_2 + 1.5x_3 \leq 20 \\ & 2x_1 + 1.5x_2 + 0.5x_3 \leq 8 \\ & x_2 \leq 5 \\ & x_i \geq 0 \; i = 1, \ldots, 3 \end{array}$$

primal solution:

$$x^\star = (2, 0, 8)$$
$$p^\star = 280$$

dual problem:

$$\begin{array}{rl} \text{minimize} & 48y_1 + 20y_2 + 8y_3 + 5y_4 \\ \text{subject to} & 8y_1 + 4y_2 + 2y_3 \geq 60 \\ & 6y_1 + 2y_2 + 1.5y_3 + y_4 \geq 30 \\ & y_1 + 1.5y_2 + 0.5y_3 \geq 20 \\ & y_i \geq 0 \; i = 1, \ldots, 4 \end{array}$$

dual solution:

$$y^\star = (0, 10, 10, 0)$$
$$d^\star = 280$$

Dakota furniture: interpretation of dual variables

$y_1^\star$ and $y_4^\star$ are both 0:
$\Rightarrow$ constraints 1 and 4 of the primal are loose (not binding)
$\Rightarrow$ neither the amount of lumber nor the maximum demand for tables keep Dakota furniture from getting a higher revenue;

$y_2^\star = 10$
$\Rightarrow$ constraint 2 of the primal is binding
$\Rightarrow$ changing the RHS of this constraint by u will change the value of the objective by $uy_2^\star = 10u$
$\Rightarrow$ \$10 is the maximum Dakota should be willing to pay for an additional unit (hour) of finishing labor;

$y_3^\star = 10$
$\Rightarrow$ constraint 3 of the primal is also binding
$\Rightarrow$ changing the RHS of this constraint by u will change the value of the objective by $uy_3^\star = 10u$
$\Rightarrow$ \$10 is the maximum Dakota should be willing to pay for an additional unit (hour) of carpentry labor;

if Dakota furniture is presented with a choice to get either one more hour of finishing labor or one more hour of carpentry labor, it will be indifferent between the two because both have the same impact on the objective function of the primal problem (revenue).

General interpretation of dual variables

Dual variable (**shadow price**) for the ith constraint of an LP is the amount by which $p^\star$ is **improved** if the right-hand side of the ith constraint is **loosened** by one (provided the same variables remain basic).

- for a maximisation LP, the shadow price of the ith constraint is equal to $+$(ith optimal dual variable)
- for a minimisation LP, the shadow price of the ith constraint is equal to $-$(ith optimal dual variable)

Interpretation of shadow prices is not always straightforward:

- in most cases, a constraint represents a limit on some resource. The shadow price of a $\leq$ constraint is the maximum price you would be willing to pay for an additional unit of that resource;
- however, be very careful with interpretation of dual variables for other types of constraints: make sure your interpretation makes sense.

Complementary Slackness

Variables in one problem are complementary to constraints in the other. That is, for each variable in the Primal, there is a constraint in the Dual. For each variable in the Dual, there is a constraint in the Primal. Complementary Slackness refers to a relationship between the slackness in a primal constraint and the slackness (positivity) of the associated dual variable.

Optimality conditions

primal and dual LP

$$\begin{array}{rl} \text{maximize} & c^T x \\ \text{subject to} & Ax \leq b \\ & x \geq 0 \end{array} \qquad \begin{array}{rl} \text{minimize} & b^T y \\ \text{subject to} & A^T y \geq c \\ & y \geq 0 \end{array}$$

optimality conditions: $\tilde{x}$ and $\tilde{y}$ are primal and dual optimal if and only if

- $\tilde{x}$ is primal feasible: $A\tilde{x} \leq b$ and $\tilde{x} \geq 0$
- $\tilde{y}$ is dual feasible: $A^T\tilde{y} \geq c$ and $\tilde{y} \geq 0$
- the duality gap is zero: $c^T\tilde{x} = b^T\tilde{y}$

Complimentary slackness

Suppose $x^\star$ is primal optimal, $y^\star$ is dual optimal. From strong duality:

$$c^T x^\star \leq (y^\star)^T A x^\star \leq (y^\star)^T b = c^T x^\star$$

Hence, the two inequalities hold with equality and lead to:

$$(y^\star)^T A x^\star = (y^\star)^T b \quad \Longrightarrow \quad (y^\star)^T (b - A x^\star) = 0$$

Complimentary slackness

This leads to complementary slackness (a_i^T are the rows of A):

$y_i^\star(b_i - a_i^T x^\star) = 0$ for $i = 1, \ldots, m$

$$\begin{aligned} y_i^\star > 0 \quad &\implies \quad a_i^T x^\star = b_i, \\ b_i - a_i^T x^\star > 0 \quad &\implies \quad y_i^\star = 0. \end{aligned}$$

in other words:

$$y_i^\star = 0 \quad \text{for all } i \text{ for which} \quad \sum_{j=1}^{n} a_{ij} x_j^\star < b_i.$$

$$x_j^\star = 0 \quad \text{for all } j \text{ for which} \quad \sum_{i=1}^{m} a_{ij} y_i^\star > c_j.$$

Variants and examples

Inequality form LP and its dual

Consider an inequality form LP and its dual, where $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$:

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \geq b \end{array} \qquad \begin{array}{ll} \text{maximize} & b^T y \\ \text{subject to} & A^T y = c \\ & y \geq 0 \end{array}$$

Observe that:

- $y \in \mathbb{R}^m \Rightarrow$ there is a dual variable for every primal constraint;
- primal constraints are $\geq \Rightarrow$ dual variables $y \geq 0$;
- x is unrestricted $\Rightarrow$ dual constraints are equalities;
- the vector of coefficients of the objective function of the primal c became the right hand side of the dual constraints;
- the vector of the right hand side of the primal constraints b became the vector of coefficients of the objective function of the dual.

Standard form LP and its dual

Consider a standard form LP and its dual, where $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$:

$$\begin{aligned} \text{minimize} \quad & c^T x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned} \qquad\qquad \begin{aligned} \text{maximize} \quad & b^T y \\ \text{subject to} \quad & A^T y \leq c \end{aligned}$$

Observe that:

- $y \in \mathbb{R}^m \Rightarrow$ there is a dual variable for every primal constraint;
- primal constraints are equalities $\Rightarrow$ dual variables y are unrestricted;
- $x \geq 0 \Rightarrow$ dual constraints are $\leq$;
- the vector of coefficients of the objective function of the primal c became the right hand side of the dual constraints;
- the vector of the right hand side of the primal constraints b became the vector of coefficients of the objective function of the dual.

Relation between primal and dual variables and constraints

We can summarise our previous observations in the following table:

	minimize	maximize	
constraints	$\geq b_i$ $\leq b_i$ $= b_i$	≥ 0 ≤ 0 free	**variables**
variables	≥ 0 ≤ 0 free	$\leq c_j$ $\geq c_j$ $= c_j$	**constraints**

Example: an LP (left) and its dual (right):

$$\begin{aligned} \text{maximize} \quad & -x_1 - x_2 \\ \text{subject to} \quad & -2x_1 - x_2 \leq 4 \\ & 2x_1 - 4x_2 \geq 8 \\ & x_1 - 3x_2 \geq 7 \\ & x_1, x_2 \geq 0 \end{aligned}$$

$$\begin{aligned} \text{minimize} \quad & 4y_1 + 8y_2 + 7y_3 \\ \text{subject to} \quad & -2y_1 + 2y_2 + y_3 \geq -1 \\ & -y_1 - 4y_2 - 3y_3 \geq -1 \\ & y_1 \geq 0 \\ & y_2, y_3 \leq 0 \end{aligned}$$

General form LP and its dual: another way of putting it

Let A be a matrix with rows $\tilde{a}_i^T$ and columns a_j. Given a **primal** problem with structure shown on the left, it's **dual** is defined to be the maximisation problem shown on the right:

$$
\begin{array}{rl}
\text{minimize} & c^T x \\
\text{subject to} & \tilde{a}_i^T x \geq b_i, \quad i \in M_1 \\
& \tilde{a}_i^T x \leq b_i, \quad i \in M_2 \\
& \tilde{a}_i^T x = b_i, \quad i \in M_3 \\
& x_j \geq 0, \quad j \in N_1 \\
& x_j \leq 0, \quad j \in N_2 \\
& x_j \text{ free}, \quad j \in N_3
\end{array}
\qquad
\begin{array}{rl}
\text{maximize} & y^T b \\
\text{subject to} & y_i \geq 0, \quad i \in M_1 \\
& y_i \leq 0, \quad i \in M_2 \\
& y_i \text{ free}, \quad i \in M_3 \\
& y^T a_j \leq c_j, \quad j \in N_1 \\
& y^T a_j \geq c_j, \quad j \in N_2 \\
& y^T a_j = c_j, \quad j \in N_3
\end{array}
$$

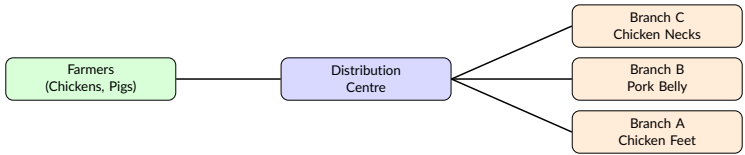
note:

M_1, M_2, M_3, N_1, N_2, and N_3 are sets of indices that correspond to various types of variables and constraints.

Sensitivity Analysis

We study the dependence of the optimal cost and the optimal solutions on the coefficient matrix A, the requirement vector b, and the cost vector c. This is an important issue in practice because we often have incomplete knowledge of the problem data and we may wish to predict the effects of certain parameter changes.

Business Motivation: Commitment Before Full Knowledge



Companies often have to commit resources months in advance — long before actual demand and prices are known with certainty. Take a supermarket chain as example:

- A major supermarket chain asked us to help with an optimization problem involving meat supply planning.
- Different suburbs prefer different cuts: chicken feet vs chicken necks, pork belly vs pork insides, etc.
- They need to place orders with farmers months in advance, since it takes time to raise chickens or pigs.
- Once the livestock is committed, the cuts cannot be rebalanced easily.

Business Motivation: Commitment Before Full Knowledge

The Optimisation Problem

Decision variables: how much of each cut to order for each branch.
Constraints:

- Total supply must match what farmers can produce.
- Each branch requires a minimum of certain cuts (local preferences).

Objective: maximize overall profit (revenue minus cost).

Business Motivation: Commitment Before Full Knowledge

Why Sensitivity Analysis Even after the optimization, uncertainty remains:

- What if demand shifts and one suburb needs more chicken feet than forecast?
- What if a farmer raises the price of pork by 5
- What if supply of one cut is unexpectedly reduced?

The supermarket doesn't want to re-solve the whole massive LP from scratch every time a parameter changes slightly. Instead, they need:

- RHS sensitivity: How much can branch demand change before the current solution breaks?
- Dual prices: If demand increases by 1 unit, how much will that change the total profit?
- Range analysis: How low can they price an item and still make a profit?

Sensitivity analysis

Purpose: extract from the solution of an LP information about the sensitivity of the solution to changes in problem data

Types of perturbations:

- changing the right-hand side of a constraint
- changing the objective function coefficient of a nonbasic variable
- changing the objective function coefficient of a basic variable

Changing the right-hand side of a constraint

(unperturbed) normal form LP and its dual

$$\begin{aligned} \text{maximize} \quad & c^T x \\ \text{subject to} \quad & Ax \leq b \\ & x \geq 0 \end{aligned} \qquad \begin{aligned} \text{minimize} \quad & b^T y \\ \text{subject to} \quad & A^T y \geq c \\ & y \geq 0 \end{aligned}$$

perturbed normal form LP and its dual

$$\begin{aligned} \text{maximize} \quad & c^T x \\ \text{subject to} \quad & Ax \leq b + u \\ & x \geq 0 \end{aligned} \qquad \begin{aligned} \text{minimize} \quad & b^T y + u^T y \\ \text{subject to} \quad & A^T y \geq c \\ & y \geq 0 \end{aligned}$$

Global sensitivity result

Assume that $y^\star$ is dual optimal for unperturbed problem.
Apply weak duality to perturbed problem:

$$p^\star(u) \leq b^T y^\star + u^T y^\star$$

Sensitivity interpretation:

- if $y_i^\star$ is large: $p^\star$ decreases greatly if we tighten constraint i $(u_i < 0)$
- if $y_i^\star$ is large: $p^\star$ increases greatly if we loosen constraint i $(u_i > 0)$
- if $y_i^\star$ is small: $p^\star$ doesn't decrease much if we tighten constraint i $(u_i < 0)$
- if $y_i^\star$ is small: $p^\star$ doesn't increase much if we loosen constraint i $(u_i > 0)$

Local sensitivity result

Suppose now that the perturbation vector u is small such that the same basic feasible solution is optimal for the perturbed problem.

This implies that the optimal objective value in the perturbed problem is

$$p^\star(u) = b^T y^\star + u^T y^\star = p^\star(0) + u^T y^\star$$

Therefore, a small change of u to the right-hand side vector b results in a change of $u^T y^\star$ in the objective value.

Hence, each component y_i of the optimal dual vector can be interpreted as the **marginal cost** (or **shadow price**) per unit increase of the ith component of b.

Changing the objective function coefficient of a nonbasic variable - Example: Dakota furniture

$$
\begin{array}{lrrrrl}
\text{maximize} & 60x_1 + & 30x_2 + & 20x_3 & & \\
\text{subject to} & 8x_1 + & 6x_2 + & x_3 & \leq & 48 \\
& 4x_1 + & 2x_2 + & 1.5x_3 & \leq & 20 \\
& 2x_1 + & 1.5x_2 + & 0.5x_3 & \leq & 8 \\
& & x_2 & & \leq & 5 \\
& x_1, & x_2, & x_3 & \geq & 0
\end{array}
$$

is equivalent to:

$$
\begin{array}{lrrrrrrrrrl}
\text{maximize} & 60x_1 + & 30x_2 + & 20x_3 & & & & & & & \\
\text{subject to} & 8x_1 + & 6x_2 + & x_3 + & s_1 & & & & & = & 48 \\
& 4x_1 + & 2x_2 + & 1.5x_3 & + & s_2 & & & & = & 20 \\
& 2x_1 + & 1.5x_2 + & 0.5x_3 & & & + & s_3 & & = & 8 \\
& & x_2 & & & & & & + \; s_4 & = & 5 \\
& x_1, & x_2, & x_3, & s_1, & s_2, & & s_3, & s_4 & \geq & 0
\end{array}
$$

Example: initial and final simplex tableau

Table: initial simplex tableau

	z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	rhs	BV
row 0	1	-60	-30	-20	0	0	0	0	0	$z = 0$
row 1	0	8	6	1	1	0	0	0	48	$s_1 = 48$
row 2	0	4	2	1.5	0	1	0	0	20	$s_2 = 20$
row 3	0	2	1.5	0.5	0	0	1	0	8	$s_3 = 8$
row 4	0	0	1	0	0	0	0	1	5	$s_4 = 5$

Table: final simplex tableau

	z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	rhs	BV
row 0	1	0	5	0	0	10	10	0	280	$z = 280$
row 1	0	0	-2	0	1	2	-8	0	24	$s_1 = 24$
row 2	0	0	-2	1	0	2	-4	0	8	$x_3 = 8$
row 3	0	1	1.25	0	0	-0.5	1.5	0	2	$x_1 = 2$
row 4	0	0	1	0	0	0	0	1	5	$s_4 = 5$

Changing the objective function coefficient of a nonbasic variable

Consider the question: how much the objective-function coefficients of non-basic variables can vary without changing the values of the decision variables in the optimal solution?

To answer that, suppose that we increase the objective function coefficient of x_2 in the original problem by Δc_2, giving us:

$$z - 60x_1 - (30 + \Delta c_2)x_2 - 20x_3 = 0.$$

Changing the objective function coefficient of a nonbasic variable

This leads to the following modified final tableau

Table: final simplex tableau

	z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	rhs	BV
row 0	1	0	5-Δc_2	0	0	10	10	0	280	z = 280
row 1	0	0	-2	0	1	2	-8	0	24	s_1 = 24
row 2	0	0	-2	1	0	2	-4	0	8	x_3 = 8
row 3	0	1	1.25	0	0	-0.5	1.5	0	2	x_1 = 2
row 4	0	0	1	0	0	0	0	1	5	s_4 = 5

The optimal solution remains unchanged so long as:

$$5 - \Delta c_2 \geq 0 \quad \text{or} \quad \Delta c_2 \leq 5 \quad \Longrightarrow \quad c_2 \leq 35$$

Changing the objective function coefficient of a basic variable

What happens when the objective-function coefficient of a basic variable is varied?

To answer that, suppose that we increase the objective function coefficient of x_1 in the original problem by Δc_1, giving us:

$$z - (60 + \Delta c_1)x_1 - 30x_2 - 20x_3 = 0.$$

Changing the objective function coefficient of a basic variable

This leads to the following modified final tableau

	z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	**rhs**	**BV**
row 0	1	$-\Delta c_1$	5	0	0	10	10	0	280	$z = 280$
row 1	0	0	-2	0	1	2	-8	0	24	$s_1 = 24$
row 2	0	0	-2	1	0	2	-4	0	8	$x_3 = 8$
row 3	0	1	1.25	0	0	-0.5	1.5	0	2	$x_1 = 2$
row 4	0	0	1	0	0	0	0	1	5	$s_4 = 5$

However, the simplex method requires that the final system of equations be in canonical form with respect to the basic variables. Since the basis is to be unchanged, in order to make the coefficient of x_1 zero in the final tableau we must add Δc_1 times row 3 to row 0 in the final tableau. The result is the Tableau on the next slide.

Changing the objective function coefficient of a basic variable

This leads to the following modified final tableau:

z	x_1	x_2	x_3	s_1	s_2	s_3	s_4	**rhs**	**BV**
1	0	$5+1.25\Delta c_1$	0	0	$10-0.5\Delta c_1$	$10+1.5\Delta c_1$	0	$280+2\Delta c_1$	$z = 280$
0	0	-2	0	1	2	-8	0	24	$s_1 = 24$
0	0	-2	1	0	2	-4	0	8	$x_3 = 8$
0	1	1.25	0	0	-0.5	1.5	0	2	$x_1 = 2$
0	0	1	0	0	0	0	1	5	$s_4 = 5$

Changing the objective function coefficient of a basic variable

By the simplex optimality criterion, all the objective-function coefficients in the final tableau must be nonnegative in order to maintain the current solution. Hence, we must have:

$$\begin{cases} 5 + 1.25\Delta c_1 \geq 0 \\ 10 - 0.5\Delta c_1 \geq 0 \\ 10 + 1.5\Delta c_1 \geq 0 \end{cases} \implies \begin{cases} \Delta c_1 \geq -5/1.25 \\ \Delta c_1 \leq 10/0.5 \\ \Delta c_1 \geq -10/1.5 \end{cases}$$

which leads to

$$-4 \leq \Delta c_1 \leq 20 \quad \implies \quad 56 \leq c_1 \leq 80$$

Workshop: `lpSolveAPI`

We have now encounter situations where the decision variable is free (it can take non-positive and non-negative values). While `lpSolve` can handle this by manual variable transformation, it is good to learn another tool that can handle this naturally.
First, install the package `lpsolveAPI` using the following command:

R

```
install.packages('lpSolveAPI')
```

R code to solve the LP

```
library(lpSolveAPI)
c <- c(...)
A <- matrix(c(...),nrow=..., byrow=T)
b <- c(...)
ineq <- c(...)
res <- make.lp(nrow(A),ncol(A))
for (i in 1:ncol(A)){
set.column(res, i, A[,i])
}
lp.control(res, sense = 'min')
set.objfn(res, c)
set.constr.type(res, ineq)
set.rhs(res, b)
res
solve(res)
get.objective(res)
get.variables(res)
```

Workshop: Sensitivity Analysis Information from Modern Solvers

Let's obtain sensitivity information from the Dakota Furniture Company example.
Documentation that helps you to read the output:
`https://www.rdocumentation.org/packages/lpSolveAPI/versions/5.5.2.0-17.9/topics/get.sensitivity.rhs`

For Python Users

In the Gurobi Python interface, you can adjust the lower bound of a variable when you define the variable.
Set the `lb` argument as a Gurobi negative infinity object `-grb.GRB.INFINITY`.
If you do not include the `lb` argument when you define your variables, Gurobi will assume your variable is non-negative.

Python

```
import gurobipy as grb
M = grb.Model("example")
x = M.addVar(lb = -grb.GRB.INFINITY)
```

Sensitivity Information in Python Gurobi

Python

```
M.Pi[ind]
M.SARHSLow[ind]
M.SARHSUp[ind]
M.RC[ind]
M.SAObjUp[ind]
M.SAObjLow[ind]
```

For other solvers, please refer to the documentation of the package for sensitivity information as the syntax varies. If you need help with this, please feel free to ask Jessica.

Quick Summary

- Given a (primal) linear programming problem we can associate with it another (dual) linear programming problem, by following a set of mechanical rules.
- Each dual variable is associated with a particular primal constraint.
- Each dual constraint is associated with a particular primal variable.
- Every dual feasible vector leads to an upper bound on the optimal value of the primal problem. This is the essence of the weak duality theorem.
- The minimisation in the dual problem is then a search for the tightest such upper bound. The strong duality theorem asserts that the tightest such upper bound is equal to the optimal primal value.
- A useful relation between the optimal primal and dual solutions is provided by the complimentary slackness conditions.

Quick Summary

- Sensitivity analysis provides some (limited) extra insights without resolving the problem.
- Most model parameters (data) are estimates and are subject to uncertainty.
- Sensitivity analysis allows us to determine how “sensitive” the optimal solution is to changes in data values.
- Modern solvers produce sensitivity analysis “for free”.
- However, when data changes the best thing to do is to resolve the problem.
- There are other ways of dealing with data uncertainty: **robust optimisation** and **stochastic optimisation**.